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- **Faculté des sciences et de technologie**

✓ **Projet #9 : Provisioning avec ansible**

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Date: 13/05/2026

Objectifs du Projet :

- ✓ Automatiser le déploiement d'un serveur.

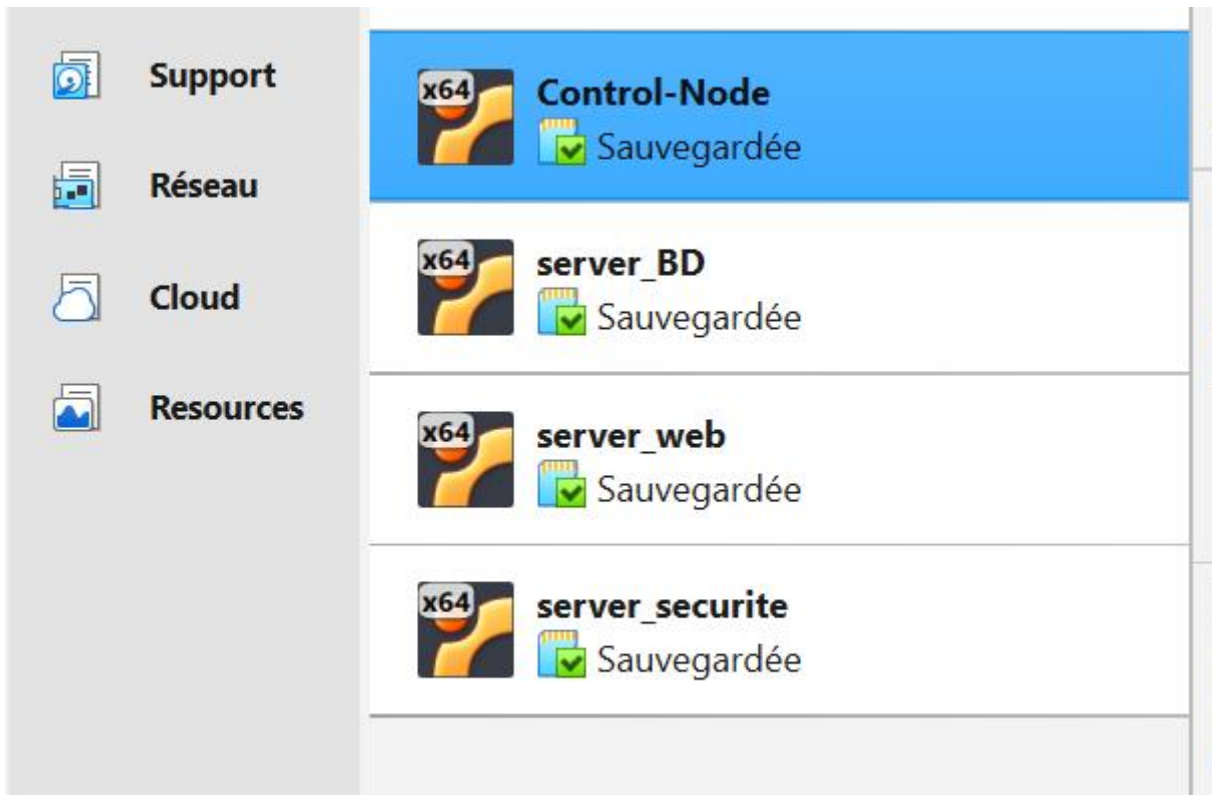
Plan du Projet :

1. Installation d'Ansible
2. Création de l'inventaire
3. Création de rôles :
 - Web
 - Base de données
 - Sécurité
4. Templates Jinja2
5. Exécution du playbook

Rapport :

- Arborescence Ansible
- Résultat de l'exécution

Tout d'abord pour réaliser ce projet je dois créer ces 4 machines



Premiere etape :1. Installer et configurer Ansible sur la machine hôte.

```
clif@ubuntuserver: ~  
clif@ubuntuserver:~$ sudo apt update  
[sudo] password for clif:  
Atteint :1 http://security.ubuntu.com/ubuntu noble-security InRelease  
Atteint :2 http://archive.ubuntu.com/ubuntu noble InRelease  
Atteint :3 http://archive.ubuntu.com/ubuntu noble-updates InRelease  
Atteint :4 https://repo.zabbix.com/zabbix/6.4/ubuntu noble InRelease  
Atteint :5 http://archive.ubuntu.com/ubuntu noble-backports InRelease  
Lecture des listes de paquets... Fait  
Construction de l'arbre des dépendances... Fait  
Lecture des informations d'état... Fait  
15 paquets peuvent être mis à jour. Exécutez « apt list --upgradable » pour les voir.  
clif@ubuntuserver:~$ sudo apt install ansible -y  
Lecture des listes de paquets... Fait  
Construction de l'arbre des dépendances... Fait  
Lecture des informations d'état... Fait  
ansible est déjà la version la plus récente (9.2.0+dfsg-0ubuntu5).  
0 mis à jour, 0 nouvellement installés, 0 à enlever et 15 non mis à jour.
```

```
clif@ubuntuuserver:~$ ansible --version
ansible [core 2.16.3]
  config file = None
  configured module search path = ['/home/clif/.ansible/plugins/modules', '/usr/share/ansible/plugins/modules']
  ansible python module location = /usr/lib/python3/dist-packages/ansible
  ansible collection location = /home/clif/.ansible/collections:/usr/share/ansible/collections
  executable location = /usr/bin/ansible
  python version = 3.12.3 (main, Mar 3 2026, 12:15:18) [GCC 13.3.0] (/usr/bin/python3)
  jinja version = 3.1.2
  libyaml = True
```

Je fais la configuration SSh entre les machines

```
clif@ubuntuuserver: ~
executable location = /usr/bin/ansible
python version = 3.12.3 (main, Mar 3 2026, 12:15:18) [GCC 13.3.0] (/usr/bin/python3)
jinja version = 3.1.2
libyaml = True
clif@ubuntuuserver:~$ ssh-keygen -t rsa
Generating public/private rsa key pair.
Enter file in which to save the key (/home/clif/.ssh/id_rsa):
/home/clif/.ssh/id_rsa already exists.
Overwrite (y/n)? y
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/clif/.ssh/id_rsa
Your public key has been saved in /home/clif/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:h/plgtEcLRf5rbWuue7H4qoLjoNcbyPCo+uLZD6ZoVo clif@ubuntuuserver
The key's randomart image is:
+---[RSA 3072]-----+
|          ..          |
|         ...         |
|        o o . .       |
|       o = . o        |
|      . S . o .       |
|     . . + . . .      |
|    .=Eo .+ . o o     |
|   ** oo+o + . .+    |
|  B*oo.+..+o.=B=     |
+-----[SHA256]-----+
clif@ubuntuuserver:~$ |
```

ici je fais la connection des machines

```
clif@ubuntuserver:~$ ssh-copy-id securite@192.168.1.229
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/clif/.ssh/id_rsa.pub"
The authenticity of host '192.168.1.229 (192.168.1.229)' can't be established.
ED25519 key fingerprint is SHA256:GXvcRBSMSmpz4wRyS8yEbf4JW0TsxkuQ1UOKj9prBoU.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
securite@192.168.1.229's password:

Number of key(s) added: 1

Now try logging into the machine, with:  "ssh 'securite@192.168.1.229'"
and check to make sure that only the key(s) you wanted were added.
```

```
clif@ubuntu:~$ ssh-copy-id server2@192.168.1.32
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/clif/.ssh/id_rsa.pub"
The authenticity of host '192.168.1.32 (192.168.1.32)' can't be established.
ED25519 key fingerprint is SHA256:WemMahaZ6x0QOHTbUi7floS12xEQDe
wMnEroMvAoeRk.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
server2@192.168.1.32's password:

Number of key(s) added: 1

Now try logging into the machine, with:  "ssh 'server2@192.168.1.32'"
and check to make sure that only the key(s) you wanted were added.
```

```
clif@ubuntuserver:~$ ssh-copy-id web@192.168.1.114
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/
home/clif/.ssh/id_rsa.pub"
The authenticity of host '192.168.1.114 (192.168.1.114)' can't b
e established.
ED25519 key fingerprint is SHA256:1JqtwHdDbt0saWjUiQ3sKRfPiVQ5Ig
kCpmW4hKP1gcI.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprin
t])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new ke
y(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- i
f you are prompted now it is to install the new keys
web@192.168.1.114's password:

Number of key(s) added: 1

Now try logging into the machine, with:  "ssh 'web@192.168.1.11
4'"
and check to make sure that only the key(s) you wanted were adde
d.
```


Ici on peut voir que toutes les connexions sont établies entre les machines

```
clif@ubuntuuserver:~$ ssh securite@192.168.1.229
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-110-generic x86_64)
```

```
* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:        https://ubuntu.com/pro
```

```
System information as of dim. 19 avril 2026 21:07:58 UTC
```

```
System load:          0.01
Usage of /:            44.7% of 11.21GB
Memory usage:         6%
Swap usage:           0%
Processes:            117
Users logged in:      1
IPv4 address for enp0s3: 192.168.1.229
IPv6 address for enp0s3: fdc8:7bb5:f446:8:a00:27ff:fece:b5f8
```

La maintenance de sécurité étendue pour Applications n'est pas activée.

32 mises à jour peuvent être appliquées immédiatement.
Pour afficher ces mises à jour supplémentaires, exécuter : `apt list --upgradable`

Activez ESM Apps pour recevoir des futures mises à jour de sécurité supplémentaires.
Visitez <https://ubuntu.com/esm> ou exécutez : `sudo pro status`


```
web × secu × servi × + v - □ ×
clif@ubuntuerver:~$ ssh web@192.168.1.114
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-110-generic x86_64)

* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:       https://ubuntu.com/pro

System information as of dim. 19 avril 2026 21:10:43 UTC

System load:          0.07
Usage of /:           44.0% of 11.21GB
Memory usage:         6%
Swap usage:           0%
Processes:            121
Users logged in:      1
IPv4 address for enp0s3: 192.168.1.114
IPv4 address for enp0s3: 192.168.1.96
IPv6 address for enp0s3: fdc8:7bb5:f446:8:a00:27ff:fe66:932

La maintenance de sécurité étendue pour Applications n'est pas activée.

32 mises à jour peuvent être appliquées immédiatement.
Pour afficher ces mises à jour supplémentaires, exécuter : apt list --upgradable

Activez ESM Apps pour recevoir des futures mises à jour de sécurité supplémentaires.
Visitez https://ubuntu.com/esm ou exécutez : sudo pro status
```

clif@ubuntuuserver:~\$ ssh server2@192.168.1.32
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-100-generic x86_64)

* Documentation: <https://help.ubuntu.com>
* Management: <https://landscape.canonical.com>
* Support: <https://ubuntu.com/pro>

System information as of dim. 19 avril 2026 21:11:27 UTC

System load:	0.0
Usage of /:	44.0% of 11.21GB
Memory usage:	6%
Swap usage:	0%
Processes:	124
Users logged in:	1
IPv4 address for enp0s3:	192.168.1.32
IPv6 address for enp0s3:	fdc8:7bb5:f446:8:a00:27ff:feca:7298

La maintenance de sécurité étendue pour Applications n'est pas activée.

94 mises à jour peuvent être appliquées immédiatement.
67 de ces mises à jour sont des mises à jour de sécurité.
Pour afficher ces mises à jour supplémentaires, exécuter : `apt list --upgradable`

Activez ESM Apps pour recevoir des futures mises à jour de sécurité supplémentaires.

Visitez <https://ubuntu.com/esm> ou exécutez : `sudo pro status`

Ici je fais la création du projet ansible

```
clif@ubuntuserver:~$ mkdir tp-ansible
clif@ubuntuserver:~$ cd tp-ansible
clif@ubuntuserver:~/tp-ansible$ nano inventory.ini
```

Création du fichier inventaire

```
clif@ubuntuserver:~/tp-ansible$ cat inventory.ini
[web]
192.168.1.114 ansible_user=web

[db]
192.168.1.32 ansible_user=server2

[security]
192.168.1.229 ansible_user=securite
clif@ubuntuserver:~/tp-ansible$ |
```

Teste de la connectivité

```
clif@ubuntuserver: ~/tp-ansil  x securite@securite: ~  x server2@server2: ~  x web@web: ~  x

clif@ubuntuserver:~/tp-ansible$ ansible all -i inventory.ini -m ping
192.168.1.114 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "ping": "pong"
}
192.168.1.32 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "ping": "pong"
}
192.168.1.229 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "ping": "pong"
}
```

Creation des roles

```
clif@ubuntuuserver:~/tp-ansible$ ansible-galaxy init roles/web
- Role roles/web was created successfully
clif@ubuntuuserver:~/tp-ansible$ ansible-galaxy init roles/db
- Role roles/db was created successfully
clif@ubuntuuserver:~/tp-ansible$ ansible-galaxy init roles/security
- Role roles/security was created successfully
clif@ubuntuuserver:~/tp-ansible$
```

Role du web

```
clif@ubuntuuserver:~/tp-ansible$ nano roles/web/tasks/main.yml
clif@ubuntuuserver:~/tp-ansible$ cat roles/web/tasks/main.yml
- name: Installer Nginx
  apt:
    name: nginx
    state: present
    update_cache: yes
  - name: Démarrer Nginx
    service:
      name: nginx
      state: started
      enabled: yes
clif@ubuntuuserver:~/tp-ansible$ |
```

```
clif@ubuntuuserver:~/tp-ansible$ nano roles/web/tasks/main.yml
clif@ubuntuuserver:~/tp-ansible$ cat roles/web/tasks/main.yml
- name: Installer Nginx
  apt:
    name: nginx
    state: present
    update_cache: yes

  - name: Démarrer Nginx
    service:
      name: nginx
      state: started
      enabled: yes

  - name: Déployer la page d'accueil
    template:
      src: index.html.j2
      dest: /var/www/html/index.html
clif@ubuntuuserver:~/tp-ansible$ |
```

Role du db

```
clif@ubuntuuserver:~/tp-ansible$ nano roles/db/tasks/main.yml
clif@ubuntuuserver:~/tp-ansible$ cat roles/db/tasks/main.yml
- name: Installer MySQL
  apt:
    name: mysql-server
    state: present
    update_cache: yes
clif@ubuntuuserver:~/tp-ansible$ |
```

Role du securite

```
clif@ubuntuuserver:~/tp-ansible$ nano roles/security/tasks/main.yml
clif@ubuntuuserver:~/tp-ansible$ cat roles/security/tasks/main.yml
- name: Installer UFW
  apt:
    name: ufw
    state: present

- name: Autoriser SSH
  ufw:
    rule: allow
    port: 22

- name: Activer le firewall
  ufw:
    state: enabled
clif@ubuntuuserver:~/tp-ansible$ |
```

Creation du Template Jinja2

```
clif@ubuntuuserver:~/tp-ansible$ nano roles/web/templates/index.html.j2
clif@ubuntuuserver:~/tp-ansible$ cat roles/web/templates/index.html.j2
<h1>Bienvenue sur {{ inventory_hostname }}</h1>
<p>Serveur déployé automatiquement avec Ansible</p>
clif@ubuntuuserver:~/tp-ansible$ |
```

Creation du playbook principal

```
clif@ubuntuuserver:~/tp-ansible$ nano site.yml
clif@ubuntuuserver:~/tp-ansible$ cat site.yml
- hosts: web
  become: yes
  roles:

- web

- hosts: db
  become: yes
  roles:

- db

- hosts: security
  become: yes
  roles:
- security
```


Execution du Playbook

```
clif@ubuntu:server: ~/tp-ansible$ ansible-playbook -i ~/tp-ansible/inventory.ini ~/tp-ansible/site.yml

PLAY [web] *****
TASK [Gathering Facts] *****
ok: [192.168.1.114]

TASK [web : Installer Nginx] *****
changed: [192.168.1.114]

TASK [web : Démarrer Nginx] *****
ok: [192.168.1.114]

PLAY [db] *****
TASK [Gathering Facts] *****
ok: [192.168.1.32]

TASK [db : Installer MySQL] *****
changed: [192.168.1.32]

PLAY [security] *****
TASK [Gathering Facts] *****
ok: [192.168.1.229]

TASK [security : Installer UFW] *****
ok: [192.168.1.229]

TASK [security : Autoriser SSH] *****
changed: [192.168.1.229]

TASK [security : Activer le firewall] *****
```

```
clif@ubuntu:server: ~/tp-ansible$ ansible-playbook -i ~/tp-ansible/inventory.ini ~/tp-ansible/site.yml

TASK [web : Démarrer Nginx] *****
ok: [192.168.1.114]

PLAY [db] *****
TASK [Gathering Facts] *****
ok: [192.168.1.32]

TASK [db : Installer MySQL] *****
changed: [192.168.1.32]

PLAY [security] *****
TASK [Gathering Facts] *****
ok: [192.168.1.229]

TASK [security : Installer UFW] *****
ok: [192.168.1.229]

TASK [security : Autoriser SSH] *****
changed: [192.168.1.229]

TASK [security : Activer le firewall] *****
ok: [192.168.1.229]

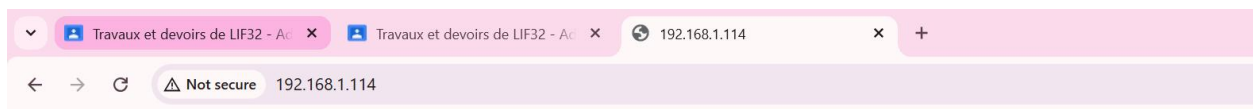
PLAY RECAP *****
192.168.1.114      : ok=3    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
192.168.1.229      : ok=4    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
192.168.1.32      : ok=2    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

clif@ubuntu:server: ~/tp-ansible$
```


Tester le serveur web :

```
clif@ubuntu-server:~/tp-ansible$ curl http://192.168.1.114
<h1>Bienvenue sur 192.168.1.114</h1>
<p>Serveur d  ploy   automatiquement avec Ansible</p>
clif@ubuntu-server:~/tp-ansible$ |
```

L'Automatisation du d  ploiement d'un serveur avec succ  s



Bienvenue sur 192.168.1.114

Serveur d  ploy   automatiquement avec Ansible